

COOLLAMP REV A – ELECTRICAL REVIEW DRAFT

NOT FOR FABRICATION – encoder bias review, protection, sourcing and footprints

1. POWER INPUT / 3V3 SUPPLY

2. CONTROLLER – FIXED GPIO MAPPING

3. USB / LED / MICROPHONE

4. HW-040 ENCODER BREAKOUT

LED count remains configurable (1-1024 software range).
Approx. 132 LEDs expected for helix; current limit is separate.
Mechanical: breakout mounting stack must be remeasured.

RELEASE BLOCKERS
HW-040 header footprint, SW bias and GPIO2 boot behaviour
Exact module footprint / antenna placement and mic sourcing
USB ESD, input protection, inrush, current limits and thermals
LED data margins; connector ratings; schematic/PCB review

HW-040: R2/R3 10k fitted; R1 appears unpopulated.
GPIO1 feeds +; CLK=GPIO4, DT=GPIO3, SW=GPIO2.
Verify resistor routing / SW bias. R6-R8 remain DNP.

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Title: CoolLamp Rev A – review schematic

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